University of Toronto Scarborough
Department of Computer and Mathematical Sciences
MATA30H3F (LEC 01\02) - Fall 2023 - Final Exam - Practice 1

Date: Tuesday, December 12, 2023 from 14:00 - 17:00
Instructor: Michael Cavers
Location(s): HL 170

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Signature: ______________________________

- **Duration:** 180 minutes
- Write your solutions in this booklet.
- This is a closed-book exam. No aids are allowed for this exam other than those provided by the instructor. Calculators and the use of personal electronic or communication devices is prohibited.
- You may write your exam in pencil, pen, or other ink. You may have an eraser, correction fluid and a ruler at your workspace.
- This exam has 17 pages with the last page being blank. There are 14 problems with the number of points indicated by each problem. The total number of points possible is 100.
- The University of Toronto's Code of Behaviour on Academic Matters (July 2019) applies to all University of Toronto Scarborough students. The Code prohibits all forms of academic dishonesty including, but not limited to, cheating, plagiarism, and the use of unauthorized aids. Students violating the Code may be subject to penalties up to and including suspension or expulsion from the University.

Instructions. Write your solutions in the space provided. Show all of your work. Full points are awarded only if your solutions are numerically correct and sufficiently display relevant concepts from MATA30.

1. (8 points) (a) Give a definition for the natural logarithm function $f(x) = \ln(x)$.

   (b) Describe what the limit notation $\lim_{x \to -\infty} f(x) = L$ means.

   (c) Give a formula for the derivative of $\dfrac{f(x)}{g(x)}$, i.e., state the quotient rule formula.

   (d) Give a precise statement of the Fundamental Theorem of Calculus. You may choose either Part I or Part II.

2. (8 points) (a) Solve for x in the equation $\ln(x) + \ln(x-1) = \ln(2)$.

(b) Find the (natural) domain of $g(x) = \cos^{-1}(2x+7)$ in interval form.

3. (7 points) Find all horizontal asymptotes of $f(x) = \dfrac{\sqrt{x^2+1}}{x}$.

4. (6 points) Find the values of a and b so that the function $f(x)$ is continuous at $x = 3$.

$$f(x) = \begin{cases} |1 - x|, & \text{if } x < 3, \\ 2b, & \text{if } x = 3, \\ \tan^{-1}(x - 2) + a, & \text{if } x > 3. \end{cases}$$

Be sure to write your solution formally using limits.

5. (6 points) A spherical balloon is being filled with air so that the radius is increasing at a rate of 2 cm/second. How fast is the volume changing when the radius is 3 cm?

   Leave your answer as a multiple of π. Note that the volume of a sphere is $V = \frac{4}{3}\pi r^3$.

6. (7 points) (a) State the Mean Value Theorem.

(b) Suppose that $3 \leq f'(x) \leq 6$ for all values of x and that $f(1) = 2$. Show that $20 \leq f(7) \leq 38$.

If you use any relevant theorems, make sure to verify their hypotheses.

7. (7 points) (a) Evaluate $\lim_{x \to 0} \frac{e^x - 1}{x}$.

(b) Evaluate $L = \lim_{x \to \infty} (e^x + x)^{1/x}$.

8. (6 points) A bacteria culture initially has 10 bacteria and in one hour has grown to 100 bacteria. Let $y(t)$ denote the amount of bacteria after t hours and assume that the bacteria culture's growth rate is proportional to its size.

   (a) Use the exponential growth/decay model to show that $y(t) = 10^{t+1}$.

   (b) How many hours does it take to reach 1000 bacteria?

   (c) How many bacteria are there after 3 hours?

9. (7 points) Find the dimensions of the rectangle with largest area that fits above the graph of the parabola $y = x^2$ and below the line $y = 3$ so that the top side of the rectangle is on the horizontal line $y = 3$.

10. (8 points) (a) Estimate the area under $f(x) = x^2$ on $[0, 3]$ by using a Riemann sum with three rectangles of equal width and choosing the right endpoints to determine the rectangle height.

(b) Using the definition for the definite integral, write $\int_0^3 x^2 \, dx$ as a limit of a Riemann sum.

You do not need to evaluate the limit.

11. (8 points) (a) Evaluate $\displaystyle\int \frac{4+\sqrt{x}}{x}\,dx$.

(b) Evaluate $\displaystyle\int_{-1}^{1} |x|\,dx$.

12. (7 points) (a) Evaluate $\int 4x^3 e^{x^4}\, dx$.

(b) Evaluate $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}}\, dx$.

13. (7 points) (a) Set up an integral that determines the area enclosed by $y = x^2$ and $y = \sqrt{x}$.

You do not need to evaluate the integral.

(b) Show that $\displaystyle\int_1^e \frac{\sin(\ln x)}{x}\,dx = 1 - \cos(1)$ by applying the substitution rule.

14. (8 points) Let $f(x) = 5 + 2x + \int_{-1}^{x^3} t^2 e^t \, dt$.

    (a) Determine all intervals where $f(x)$ is increasing.

    (b) Is $f(x)$ invertible?

    (c) Determine $f(-1)$.

    (d) Determine the value of $(f^{-1})'(3)$.

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